MATHEMATICS IN THE MODERN WORLD

Sept. 28, 2026

PART I – Direction: State whether each of the given inference is a conversion, obversion, contraposition or inversion.

1. No dogs are cats; therefore, some non-dogs are cats.
2. No fish are mammals; ergo, all fish are non-mammals.
3. Some women are not complicated; therefore, some non-complicated are women.
4. Every man is trustworthy; therefore, every untrustworthy is a non-man.
5. Some fruits are sweet; ergo, some fruits are not non sweet.
6. All luggage are heavy; therefore, some nonluggage are non-heavy.
7. All diamonds are minerals; ergo, some minerals are diamonds.
8. No wealthy person is financially insecure; therefore, some non-wealthy person are not non financially insecure.
9. Some birds are not colorful; therefore, some birds are non-colorful.
10. All superheroes are not invincible; therefore, some non-invincible are not non-superhero.
11. Some men are foreigners; therefore, some nonforeigners are non-men.
12. No fruit bearing tree is flowerless; ergo, every fruit bearing tree is nonflowerless.
13. All fishes are water creatures; ergo, no non-fish is a water creature.
14. All players are tall; therefore, all players are not small.
15. Every intelligent person is not stupid; ergo some non-stupid person are non-intelligent.
16. All dice are cubic-shaped; therefore, no noncubic-shaped are dice.
17. All pens are black; ergo, no pens are non-black.
18. All fish are sea creatures; ergo, some nonfish are not sea creatures.
19. Some teachers are considerate people; therefore, some considerate people are teachers.
20. No atheist is a believer of God; therefore, no believer of God is an atheist.

PART II – DIRECTION: Identify the given proposition and the converse as A, E, I, O. Write the converse of each of the following propositions.

1. Rizal is a province located in the Southern Tagalog region.
2. Many of the menfolk living near Laguna de Bay are fisherman.
3. Cainta is not Taytay.
4. Marikina is the town famous for shoemaking.
5. Every flower is a plant.
6. No tamarraw is a two-legged animal.
7. Filipino is the language by the people in the Philippines.
8. Hinulgang Taktak is not the falls found in San Mateo.
9. Many Belgian fathers were the parish priests of these towns.
10. Naval is the capital of Biliran.
11. No disloyal person is a true friend.
12. Some students are troublemakers.
13. Miguel is the tallest student in my English class.
14. Not all Filipinos are worth dying for.
15. Every businessman is a profiteer.
16. No frustrated person is happy.
17. An artist is a creative person.
18. Most happily married couples are compatible.
19. Some news reports are not accurate.
20. Wealthy nations are well developed.